



Model Institute of Engineering &
Technology
IEEE Student Branch



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SAMARTH

INSTANT IDEATORS

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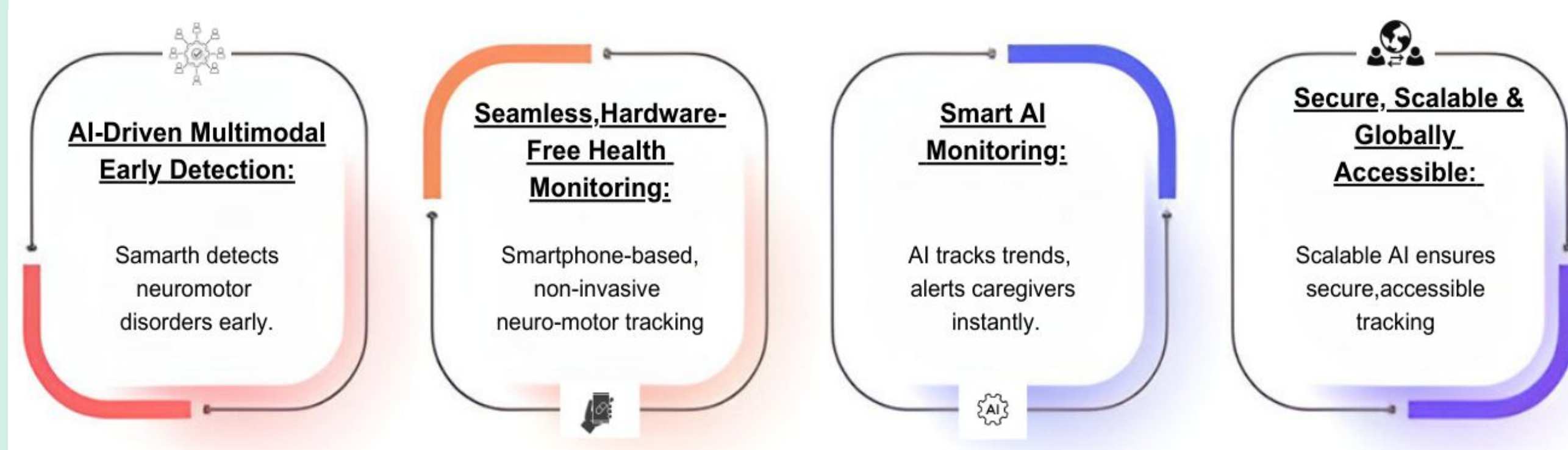
INTRODUCTION

Neuromotor disorders like **Parkinson's**, **ALS**, and **Bell's Palsy** affect millions globally and are often diagnosed too late. According to the **World Health Organization**, neurological disorders are the **leading cause of disability-adjusted life years (DALYs)** and the **second leading cause of death worldwide**. In India alone, over **7 million people** suffer from Parkinson's and related disorders, many undiagnosed due to limited access to specialists.

Traditional diagnosis relies on subjective clinical observations and expensive tests like MRI and EMG, which are often unavailable in rural areas. This creates a **critical gap in timely intervention** and continuous monitoring.

PROBLEM STATEMENT

Neuromotor disorders like **Parkinson's**, **ALS**, and **Bell's Palsy** often go undiagnosed due to infrequent check-ups and costly diagnostics. An AI-powered, real-time solution is crucial for early detection, remote monitoring, and timely intervention.

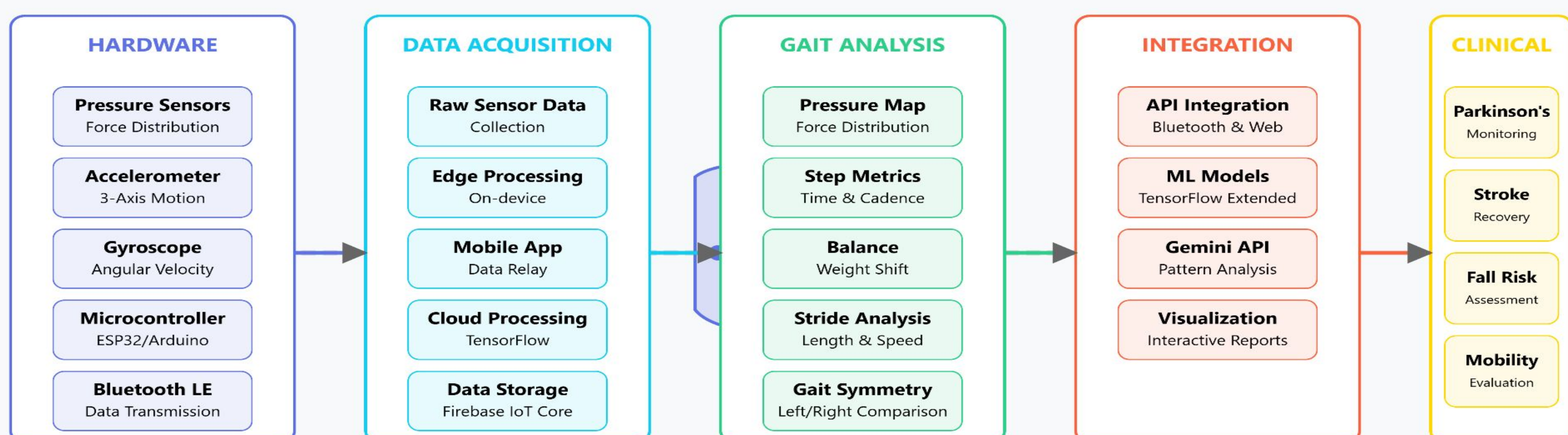


PROPOSED SOLUTION

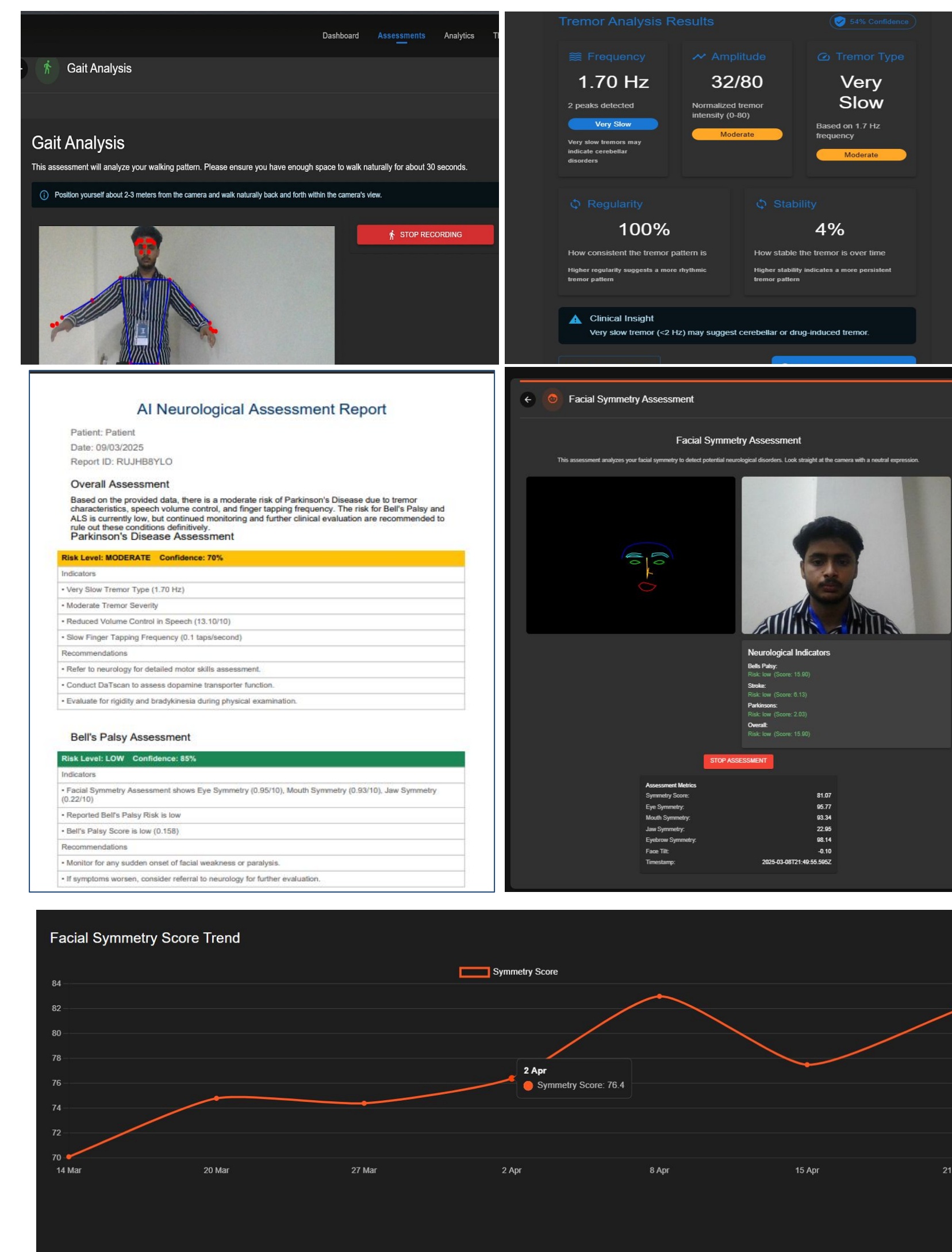
SAMARTH: AI-Powered Precision Screening & Smart Tracking for Neuromotor Health.

- Detects neuromotor disorders like Parkinson's and ALS using AI.
- Analyzes movement, facial cues, gait, voice, and tremor patterns.
- Provides real-time insights to support proactive medical decision making.
- Alerts caregivers and enables continuous, remote neurological health monitoring.

Future Development : SMART INSOLE IoT INTEGRATION



MVP Snap Shots:



ADVANTAGES & CHALLENGES

Advantages

- Early Detection:** Real-time identification of neuromotor disorders.
- Remote Monitoring:** Track patients from any location with connectivity.
- Multi-Modal Analysis:** Uses speech, gait, facial, and tremor patterns.
- Scalability:** Integrates with telehealth and public health programs.

Challenges

- Data Privacy:** Secure handling of sensitive health data.
- Connectivity:** Limited access in rural or remote areas.
- User Barriers:** Elderly may struggle with digital interfaces.
- Validation:** Needs clinical approval for diagnostic reliability.

CONCLUSION

SAMARTH is an AI-powered platform designed to **bridge the neurological care gap** by offering a **smart, scalable, and accessible** solution. It leverages advanced tools like **TensorFlow**, **OpenCV**, and **MediaPipe** to analyze complex signals such as facial expressions, gait, tremors, and speech, transforming them into **real-time health insights**.

With its **non-invasive diagnostics**, **personalized therapy engine**, and **remote monitoring capabilities**, SAMARTH enables **early detection** and **proactive care** for neuromotor disorders like Parkinson's, ALS, and Bell's Palsy. Its user-friendly interface also supports **caregivers and clinicians** with timely alerts and progress tracking.